Tanmay Bhuskute

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EDUCATION

Master of Science in Computer Science - University of Cincinnati Aug 2024 - Dec 2026
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering - AISSMS College of Engineering Aug 2018 - Jul 2022
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming & Scripting: Java, Python, JavaScript, TypeScript, C/C++, SQL, Go.
Web Backend: React, Node.js, Django, Flask, FastAPI, HTML/CSS, REST APIs.
Databases: PostgreSQL, MySQL.
Developer Tools: Git, GitLab CI/CD, Jenkins, AWS, Docker, Kubernetes, ArgoCD, Helm, Terraform, Linux, Bash, SonarQube, Azure, Google Cloud Platform.
Ai Data: Gemini, Vertex AI, PyTorch, TensorFlow, scikit-learn, LangChain, pandas, NumPy.

WORK EXPERIENCE

Siemens Industry Software (Software Development Engineer) Jul 2022 - Jun 2024
Tech Stack: AWS, Python, Boto3, REST APIs, Hashicorp Vault, Ansible, GitLab CI/CD, Shell, Kubernetes, ArgoCD, Helm, Amazon EFS

- Automated Teamcenter X infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by 40% across AWS environments
- Executed 44 production environment upgrades in 4 months, diagnosing and resolving live infrastructure issues across Linux, EC2, databases, and Teamcenter X services
- Designed Ansible-based Blue-Green deployment framework, reducing production upgrade downtime by 50% while enabling reliable rollback and recovery
- Migrated 40+ production environments from process-based deployments to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault to modernize deployment architecture
- Served as Quality Coach and SonarQube Specialist for the Agile Release Train, increasing automated code coverage from 35% to 78% by enforcing coding standards and CI/CD quality gates
- Mentored 5 software engineers through structured onboarding, technical documentation, and standardized development practices, accelerating team productivity

Government of Maharashtra, Water Resource Department (Software Developer Intern) Sep 2021 - Dec 2021
Tech Stack: React.js, MySQL, Full-Stack Development, UI/UX Design, Process Automation

- Built and deployed a full-stack React and MySQL workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by 40%

PROJECTS AND RESEARCH

PolicyPilot - Medical Benefit Policy Intelligence Portal May 2026
Tech Stack: React, TypeScript, Node.js, Express, Google Gemini, MCP, Docker, Zod, FHIR, Railway

- Architected evidence-grounded RAG pipeline over Google Gemini 2.5-flash, forcing citation-backed answers to eliminate hallucination in high-stakes healthcare eligibility decisions.
- Designed 17-tool MCP (Model Context Protocol) server exposing structured LLM tool-calling for PA extraction, patient-readiness evaluation, and cross-payer comparison, consumable by external AI agents.
- Engineered confidence-scoring system (evidence count, negation detection, subjectivity flags) driving human-review triage, migrating from Ollama to Gemini after benchmarking hallucination rates.

Memory-Augmented LLM Approach for Android Bug Reproduction from Video Reports Feb 2026 - Present
Tech Stack: Python, Google Gemini 2.5 Pro, Vertex AI, ADB, uiautomator2, scikit-image

- Proposed two-stage memory-augmented LLM pipeline for reproducing Android GUI bugs from screen recordings, decoupling video understanding from device automation to avoid repeated per-scene VLM inference.
- Evaluated against ViBR, a published VLM-based frame-accurate scene-replay approach (GPT-4o + GroundingDINO + CLIP), across 53 successful runs on 20+ real-world Android apps.
- Demonstrated 25.7x latency reduction (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity, suggesting memory reuse is a viable alternative to exhaustive scene-by-scene VLM replay.

EXTRACURRICULAR ACTIVITIES

- Finalist at RevolutionUC 2025 Hackathon, among **300 participants from 31 schools**
- Earned **SAFe Practitioner 5.1 Certification** (Mar 2023)